

C A S E S T U D Y · 2 0 2 6

MitchAI

Hospitality software prototype case study

Status & implementation boundary — What this case study does and does not claim

Status: MitchAI is a technical portfolio prototype in active development, not a public production service. Repository: <https://github.com/Bogdan0708/MitchAI> No public SLA, certification or independently assessed GDPR/compliance status is claimed.

Role: Founder and engineer, drawing on 10+ years of hospitality operations.

Author: Vasile Bogdan Godja · godjabogdan@gmail.com · github.com/Bogdan0708

01 Product scope

Hospitality workflows represented in code

Checked-in scope

- Routes and services for menus, orders, reservations, reviews, content, analytics and AI-assisted tasks.
- Authentication, tenant context, validation, rate limiting and PostgreSQL schema/migrations.
- A Next.js dashboard for hospitality and AI workflow surfaces.
- Provider adapters for cloud models and an optional local Ollama endpoint.

Boundary

Modules may require credentials, external services and further integration testing. They are implementation evidence, not evidence that every workflow runs end to end for customers.

02 Architecture

Primary runtime stated accurately

Primary stack

The backend is Node.js with TypeScript and Express. The frontend is Next.js/React, and PostgreSQL is the primary persistence layer. Redis and Qdrant are supporting services.

n8n and Ollama are optional/configured integration infrastructure. The repository does not deliver n8n workflow definitions, and Ollama requires a separate service and model. Python and FastAPI are not the primary MitchAI runtime.

Engineering posture

- Human approval remains the intended boundary for posts, messages, filings and purchases.
- Security-oriented controls are code-level signals, not certification or deployment proof.

03 Evidence & next steps

What the prototype demonstrates

What it proves

MitchAI demonstrates a broad hospitality-domain TypeScript/Express backend, Next.js frontend, PostgreSQL data model, AI-provider adapters and CI/test configuration.

Remaining gates

- Validate complete workflows against configured local dependencies.
- Publish current test/build outputs rather than fixed or historical totals.
- Complete security, privacy and deployment review before production or compliance language.